

The Collatz Conjecture at the Critical Line: Machine-Verified No-Go, Density, and Cycle Theorems

M. Sharpe*

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Abstract

We present a machine-verified account of where the Collatz conjecture is provably hard and where it is provably almost true. On the negative side, no *finite-memory descent certificate* exists: there is no potential $V(n) = nW(n)$ with W bounded above and below away from zero, no window B and contraction factor $\rho < 1$, such that every large n satisfies $V(T^t(n)) \leq \rho V(n)$ for some $t \leq B$; the obstruction is the 2-adic fixed point -1 , shadowed by the integers $2^L - 1$. On the positive side we formalize, in elementary integer arithmetic throughout: two-sided *master inequalities* $3^j(n+1) - 2^j \leq 2^T T^T(n) \leq 3^j(n+2^{T-j}) - 2^T$ for the Terras map with j odd steps among T , sharp on the shadowing orbit; the *density dichotomy* — an orbit segment avoiding descent either runs supercritical odd density $2^T < 2 \cdot 3^j$ or has exhausted the $\log_2 n$ -bit information budget of its start — together with its exact converse, so that non-descent profiles are pinned to an $O(1/B)$ -wide strip above the critical density $\log 2 / \log 3$, inhabited at every scale; the reduction of the full conjecture to universal descent; Terras’ parity-vector bijection and the binomial law $\#\{r < 2^k : j_k(r) = j\} = \binom{k}{j}$; Terras’ theorem in finite form — the integers in $[2^k, 2^{k+1})$ not descending within k steps number at most $3^k/2^{\lfloor 17(k-1)/27 \rfloor} \approx 2^{0.955k}$ — with no real analysis anywhere; and Diophantine cycle bounds excluding, conditionally on the computational verification below 2^{71} , any nontrivial cycle with at most 183 halvings, the finite check performed by the Lean kernel itself. We further report a systematic experimental program on certificates *beyond* the no-go theorem’s hypotheses — weighted automata reading all binary digits of n — identifying the complete spectrum of positive-drift rational cycles as adversaries and three mechanisms (injection, global rewrite, shift re-alignment) that empirically refute every 1- and 2-state digit certificate. All formal results are verified in Lean 4 over `mathlib` with no `sorry` and no axioms beyond standard foundations. The conjecture remains open; these results fence, with sharp constants, exactly where it lives.

1 Introduction

The Collatz function $C(n) = n/2$ for even n and $C(n) = 3n+1$ for odd n is conjectured to drive every positive integer to 1. The conjecture has been verified by exhaustive computation for all $n \leq 2^{68}$ in the published record [1], with the distributed effort since extending beyond 2^{71} , and is the subject of an extensive literature surveyed by Lagarias [7, 8].

*`msharpe@irendi.com`. The mathematics, formalization, and text of this paper were produced in collaboration with Claude (Anthropic). All formal proofs are verified by the Lean 4 kernel; see Section 12 for the audit.

Because the conjecture is so easily verified on any finite range, a perennially attractive proof strategy — attempted in countless preprints, and dissected in Section 11 — runs as follows:

1. partition the integers into finitely many classes (typically residues modulo 2^k);
2. show by finite computation that every class guarantees a decrease in some bounded number of steps, perhaps as measured by a weighted potential;
3. conclude that every large n descends below itself, and finish by strong induction into the computationally verified range.

Step 3 is sound, and step 2 *feels* like a finite computation. The first contribution of this paper is a precise and machine-verified proof that step 2 is impossible — not merely unproven, but false for every choice of class structure, potential, and window (Theorem 4.3, `no_finite_certificate`).

The second contribution is the positive complement: a formalization of exactly how close the impossible computation comes to being possible. We prove two-sided, division-free *master inequalities* for the Terras map (Section 5), from which follow: the *density dichotomy* — avoiding descent over a window of length T forces odd-step density above the critical value $\log 2 / \log 3 \approx 0.6309$, as long as the orbit is inside the $\log_2 n$ -bit information budget of its starting point — and its exact converse (Section 6): integers exist, at every scale and arbitrarily large, that avoid descent at densities within $O(1/B)$ of critical. The non-descent profiles achievable over any finite window thus form a strip of width $O(1/B)$ hugging the critical line from above, *inhabited at every scale*. Since the conjecture is equivalent to universal descent (Theorem 5.3, `collatz_iff_descent`), it is exactly the assertion that no single integer tracks this vanishing strip at every scale simultaneously — an assertion the no-go theorem shows no bounded-memory argument can decide.

The third contribution quantifies how rare the strip’s inhabitants are. We formalize Terras’ parity-vector bijection — the first k parities of an orbit are determined by, and determine, $n \bmod 2^k$ — with its corollaries that *every* parity pattern is realized and that exactly $\binom{k}{j}$ residues modulo 2^k take j odd steps (Section 7). Combining the binomial law with the dichotomy yields Terras’ theorem [13] in finite form (Section 8): the integers in the dyadic window $[2^k, 2^{k+1})$ that fail to descend within k steps number at most $3^k / 2^{\lfloor 17(k-1)/27 \rfloor} \approx 2^{0.955k}$, an exponentially vanishing fraction $\approx 2^{-0.045k}$. The proof is elementary throughout — the tail estimate is the weighted binomial theorem $2^m \sum_{j \geq m} \binom{k}{j} \leq (1+2)^k$ in plain integers, and the threshold $17/27$ comes from the single inequality $3^{17} \leq 2^{27}$ — so the formalization involves no measure theory and no real analysis. We believe this is the first machine-verified density result for the Collatz map.

Fourth, cycles (Section 9). The two master inequalities pinch on a cycle, giving the classical Diophantine constraint $n(2^T - 3^j) < 3^j 2^{T-j}$ on every cycle element, fully formal; a kernel-verified integer computation (plain `decide`; the lemma depends on the axiom `propext` alone) then shows that, conditional on the computational verification below 2^{71} — stated as an explicit hypothesis, since that computation is not formalized — no nontrivial cycle has 183 or fewer halvings.

Finally, Section 10 reports the experimental program that books-ends the formal results. The linear-programming search over joint 2-adic/3-adic residue classes, whose systematic failure motivated the no-go theorem, is complemented by two new instruments: an enumeration

of the *positive-drift rational cycles* of the Terras map — the complete spectrum of shadowing adversaries, of which the no-go theorem’s -1 is only the extreme point — and a counterexample-guided (CEGIS) search over weighted automata that read *all* binary digits of n . Such automata evade the no-go theorem’s hypotheses, and one candidate found by the search is genuinely instructive: a two-state automaton that tracks $\nu_2(n+1)$ and thereby *pre-pays* for every Mersenne rising stretch. It and all its siblings nonetheless die to three identified mechanisms — injection, global rewrite, and shift re-alignment — that we conjecture kill every finite-state digit certificate.

The Collatz conjecture remains open, and nothing in this paper proves otherwise. Section 13 states exactly what the results do *not* rule out and where the residual difficulty lives.

2 Preliminaries

For $n \in \mathbb{N}_{>0}$, the *Collatz map* is

$$C(n) = \begin{cases} n/2 & n \text{ even,} \\ 3n+1 & n \text{ odd,} \end{cases} \quad \text{and the Terras map is} \quad T(n) = \begin{cases} n/2 & n \text{ even,} \\ (3n+1)/2 & n \text{ odd.} \end{cases}$$

The two generate the same orbits up to bookkeeping: one Terras step on an odd n equals two Collatz steps, since $3n+1$ is even. We work with T because every step performs exactly one halving, which makes size estimates uniform; the formalization proves the translation (`collatz_two_eq Terras`) and states results for both maps.

For $T, n \in \mathbb{N}$ write

$$j_T(n) = \#\{t < T : T^t(n) \text{ odd}\}$$

for the number of odd steps among the first T Terras steps (`oddSteps`). The *stopping time* of n is the least t with $T^t(n) < n$, if it exists. The conjecture is equivalent to: every $n \geq 2$ has finite stopping time and the orbit of every n reaches 1; the first equivalence is made formal in Theorem 5.3.

On the 2-adic integers $\mathbb{Z}_2$, the odd branch $n \mapsto (3n+1)/2$ extends continuously, and

$$\frac{3 \cdot (-1) + 1}{2} = -1,$$

so -1 is a fixed point of the odd branch, with multiplier $3/2 > 1$ in archimedean size. No positive integer equals -1 , but positive integers approximate it 2-adically to any precision: $2^L - 1 \equiv -1 \pmod{2^L}$. More generally every rational a/q with q odd lies in $\mathbb{Z}_2$ and is approximated by integers to any 2-adic precision; this becomes important in Section 10. The next section makes the resulting shadowing exact for -1 .

3 The shadowing orbit

Lemma 3.1 (`terras_mersenne`). *For all L and all $j \leq L$,*

$$T^j(2^L - 1) = 3^j 2^{L-j} - 1.$$

Proof. Induction on j . For $j = 0$ both sides equal $2^L - 1$. For the step, let $m = 3^j 2^{L-j} - 1$ with $j < L$. Then $m = 2A - 1$ with $A = 3^j 2^{L-j-1} \geq 1$, so m is odd and

$$T(m) = \frac{3(2A - 1) + 1}{2} = 3A - 1 = 3^{j+1} 2^{L-(j+1)} - 1. \quad \square$$

In particular $T^j(2^L - 1) \geq 2^L - 1$ for all $j \leq L$ (`terras_mersenne_ge`), since $3^j 2^{L-j} \geq 2^L$; the orbit rises strictly at every step until step L . Two immediate corollaries:

Corollary 3.2 (`no_uniform_descent_bound`). *For every window B and cutoff N_0 there exists $n \geq N_0$ with $T^t(n) \geq n$ for all $t \leq B$. (Take $n = 2^{B+N_0+1} - 1$.) Equivalently, stopping times are unbounded.*

Corollary 3.3 (`uniform_descent_53_false`). *The statement “every $n > 300$ satisfies $C^t(n) < n$ for some $t \leq 53$ ” is false: $n = 2^{28} - 1$ rises for its first 56 Collatz steps.*

Corollary 3.3 is stated in this specific form as a concrete representative of the uniform-descent genre, calibrated so that no counterexample exists below the threshold (Section 11 dissects how such claims come to be “verified”). Empirically the smallest violator is $n = 447 = 110111111_2$, whose six trailing ones already exhibit the shadowing mechanism in miniature; there are 1,933 violators below 2×10^5 .

4 The no-go theorem

Definition 4.1. *A finite-memory descent certificate for T is a tuple $(W, \lambda, \Lambda, B, N_0, \rho)$ with $W: \mathbb{N} \rightarrow \mathbb{Q}$, $0 < \lambda \leq W(n) \leq \Lambda$ for all n , $B, N_0 \in \mathbb{N}$, and $0 \leq \rho < 1$, such that*

$$\forall n \geq N_0 \exists t \in [1, B]: \quad T^t(n) W(T^t(n)) \leq \rho n W(n).$$

The definition is calibrated so that a certificate would prove the conjecture:

Proposition 4.2. *If a finite-memory descent certificate exists with N_0 at most the computationally verified bound, the Collatz conjecture holds.*

Proof sketch. Set $V(n) = n W(n)$. Chaining certificate windows from any $n \geq N_0$ while the orbit remains $\geq N_0$ gives iterates n_i with $V(n_i) \leq \rho^i V(n)$, hence $n_i \leq (\Lambda/\lambda) \rho^i n \rightarrow 0$; so the orbit must drop below N_0 , where verified convergence to 1 finishes the proof. $\square$

The point of the multiplicative form is that it captures additive log-potentials with bounded correction terms. If $S: \mathbb{N} \rightarrow Q$ is *any* assignment of states in a finite set Q — a residue $n \bmod m$ for any modulus m (in particular joint 2-adic/3-adic classes $m = 2^a 3^b$), the state of any finite automaton fed by the orbit of n , any bounded window of past parities — and $\varphi: Q \rightarrow \mathbb{R}$ any weights, then $V(n) = \log n + \varphi(S(n))$ with guaranteed decrease $\varepsilon > 0$ per window is the case $W = \exp \varphi \circ S$, $\rho = e^{-\varepsilon}$, $\lambda = e^{\min \varphi}$, $\Lambda = e^{\max \varphi}$ of Definition 4.1. The unweighted descent arguments of Section 11 are the case $W \equiv 1$.

Theorem 4.3 (`no_finite_certificate`). *No finite-memory descent certificate for T exists.*

Proof. Suppose $(W, \lambda, \Lambda, B, N_0, \rho)$ were one. From the bounds, $0 < \lambda \leq \Lambda$. Choose k with $\Lambda \rho^k < \lambda$, possible since $\rho < 1$. Let $L = kB + N_0 + 1$ and $n_0 = 2^L - 1 \geq N_0$.

We chain k certificate windows inside the rising stretch of Lemma 3.1. Inductively construct times $0 = \tau_0 < \tau_1 < \dots < \tau_k$ with $\tau_i \leq iB$ and

$$T^{\tau_i}(n_0) W(T^{\tau_i}(n_0)) \leq \rho^i n_0 W(n_0). \quad (1)$$

For the step: $\tau_i \leq iB \leq kB < L$, so by Lemma 3.1 the current iterate $m_i = T^{\tau_i}(n_0)$ equals $3^{\tau_i} 2^{L-\tau_i} - 1 \geq n_0 \geq N_0$, and the certificate applies to m_i , yielding $t \in [1, B]$ with $T^t(m_i) W(T^t(m_i)) \leq \rho m_i W(m_i)$; set $\tau_{i+1} = \tau_i + t$ and combine with (1).

At $i = k$: since $\tau_k \leq kB < L$, the iterate $m_k = T^{\tau_k}(n_0)$ still satisfies $m_k \geq n_0$, whence

$$n_0 \lambda \leq m_k W(m_k) \leq \rho^k n_0 W(n_0) \leq n_0 \Lambda \rho^k < n_0 \lambda,$$

a contradiction. (Note that the certificate hypothesis was used only on integers of the single family $3^j 2^{L-j} - 1$; the theorem is thus really a statement about the orbit of $-1 \in \mathbb{Z}_2$.) $\square$

Corollary 4.4 (`no_finite_state_certificate`). *For every finite type Q , every $S: \mathbb{N} \rightarrow Q$, and every $w: Q \rightarrow \mathbb{Q}_{>0}$, there is no (B, N_0, ρ) with $\rho < 1$ such that all $n \geq N_0$ admit $t \in [1, B]$ with $T^t(n) w(S(T^t(n))) \leq \rho n w(S(n))$.*

Proof. Apply Theorem 4.3 with $W = w \circ S$, $\lambda = \min_Q w$, $\Lambda = \max_Q w$. $\square$

Remark 4.5. *The boundedness of W cannot be dropped: it is exactly what makes contraction of V assert something about descent of n (Proposition 4.2 uses both bounds). With unbounded W the quantity V can contract while n grows — already $W(n) = n^{-1-\varepsilon}$ gives a V that contracts along every rising stretch — so “certificates” in that generality must carry a validity floor $V \geq n^\delta$ to certify anything. Certificates with unbounded W satisfying such a floor are not excluded by Theorem 4.3; Section 10 investigates them experimentally. Likewise, if the window B is allowed to depend on n , the unweighted statement becomes the finiteness of stopping times, i.e. the open descent half of the conjecture itself.*

5 The master inequalities and the density corridor

Everything in this and the next four sections is elementary, exact, and division-free; the formal proofs run in $\mathbb{N}$ with no truncated subtraction. The engine is a pair of one-step identities: for odd x , $2(T(x) + 1) = 3(x + 1)$ exactly, and for even x , $2(T(x) + 1) = x + 2$. Induction along the orbit gives:

Theorem 5.1 (`master inequalities`; `terras_growth_bound`, `terras_lower_bound`). *For all T, n , with $j = j_T(n)$,*

$$3^j(n+1) - 2^j \leq 2^T T^T(n) \leq 3^j(n + 2^{T-j}) - 2^T.$$

Both bounds are achieved with equality on the shadowing orbit $n = 2^L - 1$, $T = L$ (`terras_growth_bound_sharp`), where both sides reduce to Lemma 3.1; neither can be improved. The upper bound has the following striking consequence. Call density j/T *supercritical* when $3^j > 2^{T-1}$, i.e. essentially $j/T > \log 3 \approx 0.6309$.

Theorem 5.2 (density dichotomy; `no_descent_dichotomy`). *If $n \leq T^T(n)$ then, with $j = j_T(n)$, either*

$$2^T < 2 \cdot 3^j \quad \text{or} \quad n < 2^{T-j}.$$

In words: an orbit segment that avoids descent either runs supercritical odd density, or has already spent more than $\log_2 n$ halvings. Since each halving consumes one bit of n — a heuristic made precise by the parity bijection of Section 7 — the second branch says the segment has exhausted the information content of its starting point. In particular (`early_no_descent_forces_density`), within the first $\log_2 n$ worth of halvings, survival *forces* supercritical density.

Finally, the corridor matters because descent is the whole conjecture:

Theorem 5.3 (descent reduction; `collatz_iff_descent`). *The Collatz conjecture holds if and only if every $n \geq 2$ satisfies $T^t(n) < n$ for some t .*

The forward direction tracks the Terras orbit inside the Collatz orbit; the converse is strong induction, descending through the first t with $T^t(n) < n$. Both directions are formal, so any future descent theorem converts mechanically into the full conjecture.

6 The critical line is sharp

Theorem 5.2 forbids sub-critical survival inside the information budget. The lower master inequality provides the exact converse: supercritical *prefixes* force survival.

Theorem 6.1 (`no_descent_of_supercritical_prefix`). *If $2^t \leq 3^{j_t(n)}$ for all $t \leq T$, then $n \leq T^t(n)$ for all $t \leq T$.*

Proof. By Theorem 5.1 (lower bound) and $2^j \leq 2^t \leq 3^j$: $2^t T^t(n) \geq 3^j(n+1) - 2^j \geq 2^t n + (3^j - 2^j) \geq 2^t n$. $\square$

Which prefixes occur? By the realization theorem of Section 7 (`parity_pattern_realized`), *all* of them: in particular the extremal word $1^j 0^s$ (j odd steps, then s even ones), whose every prefix is supercritical as long as $2^{j+s} \leq 3^j$. This yields adversaries at every supercritical density — the Mersenne family of Corollary 3.2 is the case $s = 0$, density 1 — and, choosing j minimal, adversaries hugging the critical line:

Theorem 6.2 (`critical_adversary`, `critical_line_sharp`). *(i) For all j, s, N_0 with $2^{j+s} \leq 3^j$ there exists $n \geq N_0$ with $j_{j+s}(n) = j$ and $n \leq T^t(n)$ for all $t \leq j+s$. (ii) For every window B and every N_0 there exists $n \geq N_0$ with $n \leq T^t(n)$ for all $t \leq B$ and*

$$2^B \leq 3^{j_B(n)} \leq 3 \cdot 2^B.$$

Part (ii) pins the odd density of an actual non-descending integer within $O(1/B)$ of $\log 2 / \log 3$ from above, for every B ; part (i) of Theorem 5.2 forbids anything below. The *non-descent profiles achievable over a window B are exactly a strip of width $O(1/B)$ above the critical line, inhabited at every scale by arbitrarily large integers*. By Theorem 5.3, the conjecture is the assertion that no single integer inhabits the strip at every scale simultaneously. (Numerically, the construction at $j = 100$ produces integers of density 0.6329, against the critical 0.6309, with no descent through all 158 steps.)

7 The parity bijection and the binomial law

The constructions above rest on Terras' structural theorem [13], which we formalize in both directions. Congruence is mod 2^k throughout.

Theorem 7.1 (`parity_of_modEq`, `modEq_of_parity`). *For all k : $n \equiv m \pmod{2^k}$ if and only if $T^t(n) \equiv T^t(m) \pmod{2}$ for all $t < k$. Consequently the map sending $r < 2^k$ to its parity vector $(T^t(r) \bmod 2)_{t < k}$ is a bijection onto $\{0, 1\}^k$.*

The proof in both directions rides on the doubling identities of Section 5: doubling carries congruence modulo 2^k to congruence modulo 2^{k+1} and back, and 3 is invertible modulo every power of 2. Three corollaries do real work:

Corollary 7.2. (i) (`oddSteps_modEq`) $j_k(\cdot)$ is a function of $n \bmod 2^k$. (ii) (`parity_pattern_realized`) Every subset of $\{0, \dots, k-1\}$ is the odd-step set of exactly one $r < 2^k$ — every finite behavior occurs. (iii) (`card_oddSteps`, the binomial law)

$$\#\{r < 2^k : j_k(r) = j\} = \binom{k}{j}.$$

Item (ii) explains the failure of worst-case certificates structurally: the adversarial pattern (all ones) is always present, exactly once, at every scale — and item (iii) says it and its supercritical neighbours are binomially rare, which is the engine of the next section.

8 Almost every integer descends: Terras' theorem, finite form

Fix k and consider the dyadic window $[2^k, 2^{k+1})$, which contains exactly one integer from each residue class modulo 2^k . If n in the window does not descend within k steps, the dichotomy (with the window position supplying $n \geq 2^k \geq 2^{k-j}$) forces supercritical density $2^k < 2 \cdot 3^{j_k(n)}$; by Corollary 7.2(i) and (iii) the eligible residues number $\sum_{j: 2^k < 2 \cdot 3^j} \binom{k}{j}$. Two elementary integer facts finish:

$$2^m \sum_{\substack{j \leq k \\ j \geq m}} \binom{k}{j} \leq \sum_{j \leq k} \binom{k}{j} 2^j = (1+2)^k = 3^k \quad (\text{choose_tail_bound}),$$

and $2^k < 2 \cdot 3^j$ implies $j \geq \lfloor 17(k-1)/27 \rfloor$, because $3^{17} \leq 2^{27}$ (`tail_exponent`; note $17/27 = 0.6296\dots$ approximates $\log 2 / \log 3 = 0.6309\dots$ from below).

Theorem 8.1 (Terras' theorem, finite form; `no_descent_window_bound`). *For every k ,*

$$2^{\lfloor 17(k-1)/27 \rfloor} \cdot \#\{n \in [2^k, 2^{k+1}) : n \leq T^t(n) \text{ for all } t \leq k\} \leq 3^k.$$

The count is therefore at most $3^k / 2^{\lfloor 17(k-1)/27 \rfloor} \approx 2^{0.955k}$: a fraction $\approx 2^{-0.045k}$ of the window, vanishing exponentially. The same bound holds for integers whose orbit *never* descends (`divergent_window_bound`): almost every integer has finite stopping time. The formal proof uses no real numbers, no measure theory, and no entropy estimates — only the weighted binomial theorem and one numeric inequality — which we found necessary as well as pleasant: the natural analytic route through the popcount of 3^j runs into a genuinely open problem

(Section 13). Numerically the truth tracks the bound comfortably: the actual fraction of non-descenders falls 0.50, 0.19, 0.063, 0.029 at $k = 1, 4, 10, 18$.

Together with Section 6: the strip above the critical line is inhabited at every scale (Theorem 6.2), but its population is an exponentially vanishing minority (Theorem 8.1). “Almost all” is now formal; “all” is the conjecture.

9 Cycle bounds

On a cycle the two master inequalities pinch against each other. Suppose $T^T(n) = n$ with $T \geq 1$, $n \geq 1$, and write $j = j_T(n)$.

Theorem 9.1 (`cycle_has_odd_step`, `cycle_three_pow_lt`, `cycle_min_bound`). $j \geq 1$; moreover $3^j < 2^T$ strictly, and

$$n \cdot 2^T < n \cdot 3^j + 3^j 2^{T-j}, \quad \text{i.e.} \quad n < \frac{3^j 2^{T-j}}{2^T - 3^j}.$$

Every cycle element is thus bounded by how well 3^j approximates 2^T from below — the classical constraint [3, 10], now fully formal. The bound has teeth at small T because good rational approximations to $\log_2 3$ are sparse. A finite computation — performed by the Lean kernel itself via plain `decide`, with no `native_decide` and no trusted code; the resulting lemma `cycle_check_183` depends on the axiom `propext` alone — verifies that for every $T \leq 183$ and every admissible j ,

$$3^j 2^{T-j} \leq (2^T - 3^j) \cdot 2^{71},$$

so every element of such a cycle is below 2^{71} .

Theorem 9.2 (`no_small_terras_cycle`). Assume every $0 < m < 2^{71}$ reaches 1 under the Collatz map (the computational verification [1], stated as a hypothesis). Then every $n \geq 1$ with $T^T(n) = n$ for some $1 \leq T \leq 183$ reaches 1; equivalently, no nontrivial cycle has 183 or fewer halvings.

The constant is sharp for this method at this verification bound: at $(T, j) = (184, 116)$ the element bound first clears the verified range $(3^{116} 2^{68} / (2^{184} - 3^{116}) \approx 2^{71.25})$, the shadow of the approximation $116 \cdot \log_2 3 \approx 183.86$. Pushing further requires a larger verified range or convergent-by-convergent Diophantine input [3], not new ideas — which is itself consistent with Theorem 4.3: cycle exclusion is finite-information reasoning, and provably cannot close the conjecture.

10 Certificate searches: from residue classes to digit automata

10.1 The residue-class search

Theorem 4.3 was found by attempting its negation. For state spaces $Q = \mathbb{Z}/2^a \times \mathbb{Z}/3^b$ one can search for additive certificates $\varphi: Q \rightarrow \mathbb{R}$ mechanically. Model the dynamics as a finite graph on Q : from a state (r_2, r_3) , the next T -iterate’s class is determined except for one bit — each halving consumes one bit of 2-adic information — so each state carries two out-edges (the 3-adic part is exact, 2 being invertible modulo 3^b), with weight $-\log 2$ on even states and

$+\log \frac{3}{2}$ on odd states. A certificate exists if and only if the associated linear program is feasible, equivalently (by LP duality) if and only if every cycle of this graph has negative mean weight, which Karp’s algorithm [4] decides exactly.

The graph over-approximates the true dynamics (both lifts of the lost bit are kept), which is sound for worst-case certificates; and the blocking cycle found below is realized by genuine integers, so no slack is lost. The results:

(a, b)	states	max mean cycle weight	verdict
(4, 0)	16	+0.405465	infeasible
(6, 1)	192	+0.405465	infeasible
(6, 2)	576	+0.405465	infeasible
(5, 3)	864	+0.405465	infeasible
(8, 1)	768	+0.405465	infeasible

The maximum mean cycle weight equals $\log \frac{3}{2} = 0.405465\dots$ exactly, at every modulus, and the optimal cycle is always the same: the self-loop at $n \equiv -1 \pmod{2^a}$. This is the 2-adic fixed point of Section 3 seen at finite resolution, and tracing it back to the integers $2^L - 1$ converts the failed search into Theorem 4.3. The search tool (`analysis/certificate_search.py`) remains useful as a mechanical refuter: any future claimed finite-modulus certificate must, by the theorem, contain an error that the tool will locate as a positive-mean cycle.

10.2 The adversary spectrum: positive-drift rational cycles

The Terras map preserves odd denominators: $T(a/q)$ is $(a/2)/q$ or $((3a + q)/2)/q$. Every rational $a/q \in \mathbb{Z}_2$ is shadowed by integers $n \equiv aq^{-1} \pmod{2^L}$ for $\sim L$ steps, so every T -cycle of rationals with *positive drift* ($3^j > 2^P$ over the cycle) is an adversary family in the sense of Section 4. Enumerating all cycles with $q \leq 99$ odd and starting numerators $|a| \leq 6q$ (`analysis/rational_cycles.py`) finds 16 positive-drift cycles in that range, all on negative rationals; the strongest:

cycle through	period P	odd steps j	density	drift ($\log_2/\text{step}$)
-1	1	1	1.000	+0.585
-65/49	5	4	0.800	+0.268
-19/11	4	3	0.750	+0.189
-5	3	2	0.667	+0.057
-17	11	7	0.636	+0.009

The no-go theorem’s -1 is the extreme point of a structured spectrum whose densities descend toward the critical line — the dynamical counterpart of Theorem 6.2.

10.3 Beyond bounded memory: digit automata and three killers

By Remark 4.5, certificates with *unbounded* W are not excluded, provided they carry a validity floor $V(n) \geq n^\delta$. The natural such family: weighted automata reading all binary digits of n (least-significant first), $\log_2 V(n) = \log_2 n + \sum_i g(q_i, b_i)$, which can vary over $n^{\Theta(1)}$ scales. We searched this family systematically (`analysis/automaton_potentials.py`,

`analysis/transducer_search.py`); since $\log V$ is linear in the weight vector g , fitting g against an adversary pool is a piecewise-linear min-max problem, and we run full counterexample-guided synthesis: fit, attack (Mersenne family, rational-cycle shadows, exact minimal-weight strings by dynamic programming, shift injections, hill-climbing over bit flips), add the violator to the pool, repeat.

One-state results. The popcount potential $V(n) = n \cdot c^{\text{ones}(n)}$ contracts along the Mersenne orbit precisely when $c > 9/4$ — the measured boundary $\log_2 c = 1.17$ matches the prediction from $\text{ones}(3^j 2^{L-j} - 1) \approx L - j/2$ — so it kills the no-go adversary. It dies elsewhere: the full two-parameter family $V(n) = n 2^{g_1 \text{ones}(n) + g_0 \text{zeros}(n)}$ is refuted across the entire (g_0, g_1) plane.

Two-state results. All 16 two-state transition structures, with the validity floor enforced as linear constraints (every cycle of the weight graph must have mean $\geq -(1 - \delta)$; without it the optimizer returns inverted potentials $V \sim n^{-\gamma}$ that certify nothing), are refuted by the CEGIS loop. The most instructive candidate is the *trailing-run automaton*, whose state records whether the read head is still inside the initial run of ones — so it computes $\nu_2(n+1)$, the exact predictor of Mersenne rising stretches — with fitted potential $V(n) \approx n^{1.79} \cdot 4.6^{\nu_2(n+1)}$. It *pre-pays* for every rise and genuinely contracts along the entire Mersenne family. It dies in one step: for $n = 2(2^{40} - 1)$, the first halving re-aligns the read head with the giant run of ones and releases the stored weight at once (measured $+86.8$ in $\log_2 V$; the 30-step window minimum stays at $+52.9$).

Three refutation mechanisms, each with constructive adversaries, emerged:

1. **Injection:** strings of extreme low weight (computable exactly by dynamic programming on the automaton) whose statistics the dynamics refills — the orbit may even *descend* while the potential explodes (a measured case: $\Delta \log_2 n = -8.3$ over 40 steps while $\Delta \log_2 V = +56$, the ones-density recovering from 0.025 toward typical).
2. **Global rewrite:** the odd step multiplies the entire remaining bit string by 3; digit statistics are not locally controlled, and even positive-drift cycle shadows pump popcount-style weights.
3. **Shift re-alignment:** halvings shift the tape, so any LSB-anchored state has its stored potential released by a single even step, as above.

Mechanism 3 is evaded exactly by *shift-coherent* weights — sliding k -gram statistics — whose $k = 1, 2$ cases are killed by mechanisms 1–2. We record the resulting conjecture:

Conjecture 10.1. *No k -gram digit potential (for any k) with validity floor $V \geq n^\delta$, window B , and factor $\rho < 1$ is a descent certificate for T .*

Base-2 statistics die on $\times 3$; base-3 statistics die, symmetrically, on halving. A weight statistic compatible with both must live on the joint 2-adic/3-adic structure — the $\times 2, \times 3$ rigidity landscape of Furstenberg’s times-two, times-three problem — which we take as quantitative evidence that the difficulty of the conjecture is located exactly where the folklore places it. These are experimental refutations: every kill is an explicit, checkable violator, but “no certificate exists in this family” remains conjecture pending a proof of mechanism 1 or 3 in the style of Section 6 (prescribed parity words rather than explicit bit strings; see Section 13).

11 Case study: how claims like Corollary 3.3’s negation get “verified”

Arguments of the genre formalized in Definition 4.1 typically proceed by tracking, for each odd residue class modulo 2^{14} (say), an affine form $c \cdot n + d$ along the orbit, with worst-case handling of boundary cases, and reporting that every class descends within some bound — 53 steps, for the claim refuted in Corollary 3.3. We implemented exactly such a tracker to locate the characteristic flaw. It is this: the class of n modulo 2^{14} determines the orbit’s parity decisions only until the cumulative number of halvings reaches 14 — each halving spends one bit, exactly as in Theorem 7.1 — while the tracker runs for up to 53 steps. Beyond the budget, the tracked trajectory silently follows the class representative rather than the class. For the class of $n = 10087$, the tracker first diverges from reality at step 12 (cumulative halvings 15), then certifies a descent that $n = 10087$ itself violates: its actual stopping time is 66 Terras steps.

We record this because it is the *typical* error of the genre, and because Theorem 4.3 explains its inevitability: had the precision accounting been done correctly at modulus 2^{14} , the verification would have failed on the classes near -1 ; enlarging the modulus relocates but never removes them. By the binomial law (Corollary 7.2) the classes with no guaranteed early descent number $\sum_{j>0.63k} \binom{k}{j} \sim 2^{0.95k}$ at modulus 2^k : their density tends to 0 — that is Theorem 8.1 — but their count grows exponentially, which is why “almost every” is where this method provably stops.

12 The formalization

All theorems of Sections 3–9 are formalized in Lean 4 over mathlib [9], about 2,100 lines across seven files:

file	contents	headline results
Basic.lean	maps, 2-adic valuation	CollatzConjecture
NoGo.lean	§3–4	no_finite_certificate
Density.lean	§5	no_descent_dichotomy, collatz_iff_descent
Cycles.lean	§9	cycle_min_bound, no_small_terras_cycle
Parity.lean	§7	card_oddSteps, parity_pattern_realized
Terras.lean	§8	no_descent_window_bound
Critical.lean	§6	critical_line_sharp

The no-go statement, as type-checked:

```

theorem no_finite_certificate
  (W : ℕ → ℚ) (lo hi : ℚ) (hlo : 0 < lo)
  (hW : ∀ n, lo ≤ W n ∧ W n ≤ hi)
  (B N₀ : ℕ) (ρ : ℚ) (hρ₀ : 0 ≤ ρ) (hρ : ρ < 1)
  (hcrt : ∀ n, N₀ ≤ n → ∃ t, 1 ≤ t ∧ t ≤ B ∧
    (terras_iter t n : ℚ) * W (terras_iter t n)
      ≤ (n : ℚ) * W n * ρ) :
  False

```

and Terras’ theorem in finite form:

```
theorem no_descent_window_bound (k : ℕ) :
  2 ^ (17 * (k - 1) / 27) *
    ((Finset.Ico (2 ^ k) (2 ^ (k + 1))).filter
      (fun n => ∀ t, t ≤ k → n ≤ terras_iter t n)).card
  ≤ 3 ^ k
```

Audit. Beyond a successful clean build, we verified: (i) `#print axioms` on the 24 headline theorems reports only `propext`, `Classical.choice`, and `Quot.sound` — Lean’s standard foundations; the kernel computation `cycle_check_183` depends on `propext` alone; (ii) the sources contain no `sorry`, no axiom declarations, no `unsafe` or `partial` definitions, no `@[implemented_by]` and no `native_decide`, so no result depends on compiled code or escape hatches; (iii) every quantitative statement was cross-checked against independent Python implementations: the master inequalities on 20,000 random (n, T) , the dichotomy on all non-descending cases among them, the binomial law exhaustively for $k \leq 12$, the window bound and actual survivor counts for $k \leq 18$, the critical adversaries to $j = 100$, the sharpness identity at $L \in \{5, 10, 20, 28\}$, and the $528,958,107,647 = T^{17}(2^{29} - 1)$ spot check of Lemma 3.1; and (iv) the only hypothesis anywhere that is not formally proved is the explicitly displayed verification hypothesis of Theorem 9.2.

We are aware of formalizations of Collatz *computations* and of the conjecture’s statement in several proof assistants; we are not aware of a prior machine-verified impossibility theorem for a class of Collatz proof strategies, nor of a prior machine-verified density result for the map (Theorem 8.1), though we would welcome correction.

13 What remains

The formal results fence off worst-case bounded-memory arguments (Theorem 4.3), and the experiments of Section 10.3 push the fence, so far without exception, through the unbounded digit-automaton families. What remains available:

1. **Density and averaging arguments.** These carry unbounded information and are untouched; indeed Section 8 is one, now formal, and Tao’s result that almost all orbits attain almost bounded values [12] is their modern summit. But they prove “almost all,” and Theorem 6.2 shows the exceptional strip they permit is genuinely inhabited at every finite scale.
2. **Cycle exclusions** (Section 9): real theorems, conditional only on verified ranges, but provably finite-information — they bound cycle lengths, never settle divergence.
3. **Joint 2-adic/3-adic structure.** The three killers of Section 10.3 leave exactly one road into certificate land: statistics quasi-invariant under both $\times 3$ -with-carry and halving, i.e. living on $\mathbb{Z}_2 \times \mathbb{Z}_3$. This is the times-two/times-three rigidity landscape, and we read the experiments as quantitative evidence that the conjecture’s difficulty is exactly the difficulty of that landscape.
4. **Genuinely new ideas** exploiting the specific arithmetic of 2 and 3. Some constraint of this kind is mandatory: the generalized Collatz problem is undecidable [2, 6], so no argument indifferent to the particular map can succeed.

Two concrete formal targets suggest themselves. First, Conjecture 10.1: the injection mechanism is constructive, and the realization machinery of Section 7 can prescribe parity words without controlling explicit bit strings — a route that matters because the naive analysis of even the popcount potential on the Mersenne orbit needs the popcount of 3^j , on which nothing beyond Stewart-type logarithmic bounds is known [11]; any proof must go around that open problem, and our finite-form Terras proof shows such detours exist. Second, the quantitative no-go: certificates with window $B(n) = o(\log n)$ should be excludable by the chaining of Theorem 4.3 run inside Theorem 6.2’s adversaries.

The conjecture (descent part) is equivalent, with sharp constants on both sides now formal, to the assertion that no orbit tracks the critical strip $2^T \leq 3^{j_T(n)} \leq 3 \cdot 2^T$ at every scale T . Random parities give density $\frac{1}{2}$, far below critical; the binomial law makes the strip exponentially thin; the realization theorem keeps it nonempty at every scale; and undecidability of the generalized problem warns that only the specific arithmetic of 2 and 3 can finish. We see this paper as a fence around the parts of the search space that are provably or demonstrably empty — with the gate, the critical line, now surveyed exactly — and as a case study in making impossibility *informative*.

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Reproducibility

All Lean sources, the analysis scripts (deterministic seeds), and build instructions are in the repository accompanying this paper. `lake build` checks every theorem; `#print axioms` reproduces the audit; each script regenerates the corresponding table or experiment of Sections 10 in minutes on commodity hardware.

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